

Zeus Documentation



Southampton University Electric Propulsion

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Nomenclature

ϵ_0	Vacuum Permittivity
λ_D	Debye Length
λ_i	Ionisation Free Path
ν_i	Ionisation Frequency
σ_i	Ionisation Cross Section
k_B	Boltzmann Constant
L_c	Channel Length
m_n	Neutral Particle Mass
n_e	Electron Number Density
n_n	Neutral Number Density
T_e	Electron Temperature
T_n	Neutral Gas Temperature
v_e	Ionising Electron Velocity
$\dot{m}$	Mass flow rate
e	Elementary charge of an electron
LME	Liquid Metal Embrittlement
m_e	Mass of an electron
m_i	Mass of an ion
P_{vap}	Vapour Pressure
PDS	Propellant Delivery System
T_{cruc}	Crucible temperature

Chapter 1

Thruster

Chapters/Thruster

1.1 Requirements Definition

The thruster is designed to take the gaseous propellant, ionise it and accelerate it out of the channel. The thruster will be optimised for zinc but will still be capable of running on xenon and krypton. The selected design parameters are highlighted below in Table ???. The table separates inputs from performance parameters, which are calculated from the standard performance equations, to show the nominal upper limits of the system.

$$v_{ex} = \sqrt{\frac{2qV_d}{m_i}} \quad (1.1)$$

$$T = \dot{m}_a v_{ex} = I_d \sqrt{\frac{2m_i V_d}{q}} \quad (1.2)$$

Where v_{ex} is the exhaust velocity, q is charge, V_d is the discharge voltage, m_i is the ion mass, T is thrust, I_d is discharge current, and $\dot{m}_a$ is the mass flow rate. Equation 1.1 and 1.2 assume that there are no losses and a 100% anode efficiency.

Table 1.1: The input design parameters to be used in scaling the size of the thruster, along with the nominal performance.

Design Parameter	Value
Nominal Anode Power	500W
Power Range	(100 – 1000)W
Nominal Anode Voltage	200V
Voltage Range	(100 – 600)V
Optimum Propellant	Zinc
Magnetic Field Strength Range	(100 – 300)G
Specific Impulse	2477s
Thrust	41.2mN
Mass Flow Rate	1.69mg/s or 34.8sccm

The design parameters are selected in this manner to keep the system small so that it can be built on a modest budget while also maintaining relevance. It is also an operating region which has test data but no known optimised thruster was found for. Below is a list of high-level requirements that were set out at the start of the design.

- The thruster must operate in the operating range highlighted in Table 1.1.
- The thruster must produce a uniform gas discharge throughout the channel.
- The thruster must produce a magnetically shielded configuration to protect the discharge channel.
- The thruster must be able to operate at a steady state condition, without malfunction.
- The thruster must be able to operate on a range of gaseous propellants, but should be optimised for zinc.

1.2 Discharge Channel

A hall effect thrusters design should start with the channel geometry of which the purpose is to ionise and accelerate the propellant. This started with designing a large database of Hall effect thrusters, of which currently contains 54 thrusters and close to 1000 data points. The data is then used in 1.2.1 to determine the cross-sectional geometry of the channel for the HET. It will also be used later for verifying operating conditions and to reference performance from other thrusters.

1.2.1 Scaling Law Analysis

Standard scaling laws have been used to analyse the database to size the zinc thruster channel. The scaling laws operate under the following assumptions:

- All quantities are steady in time and uniform in space.

- The electron temperature is unchanging whatever the operating conditions.
- The propellant gas has a uniform and fixed temperature all over the channel, hence a constant propellant velocity.
- The potential energy is fully converted into kinetic energy and all ions experience the whole potential drop.
- Plasma-wall interactions are taken into account through heat load to the channel walls.
- The magnetic field is uniform; solely its value at the exit plane is considered. The channel length L is therefore the length of the region with a magnetic field.
- Electron transport across the magnetic barrier is considered as classical: no anomalous transport is accounted for within the region of strong magnetic field.
- There are no multiply charged ions in the plasma.
- A parallel monokinetic ion beam is produced, i.e., the plasma jet divergence is null ($\theta_d = 1$).

The two important equations derived by Dannenmeyer et al.(1) were used to scale the mean channel diameter and channel width from the power inputs.

$$P = C_p d^2 V_d \quad (1.3)$$

$$h = C_{hd} d \quad (1.4)$$

P is power, C_P is the power scaling coefficient, d is the mean channel diameter where $d = \frac{1}{2}(d_o + d_i)$, d_o is the outside channel diameter, d_i is the inside channel diameter, V_d is the discharge voltage, h is the channel width, and C_{hd} is the channel width scaling coefficient.

By plotting graphs of Equation 1.3 and 1.4 for every propellant from the database, we can determine the C_P and C_{hd} respectively from the gradients.

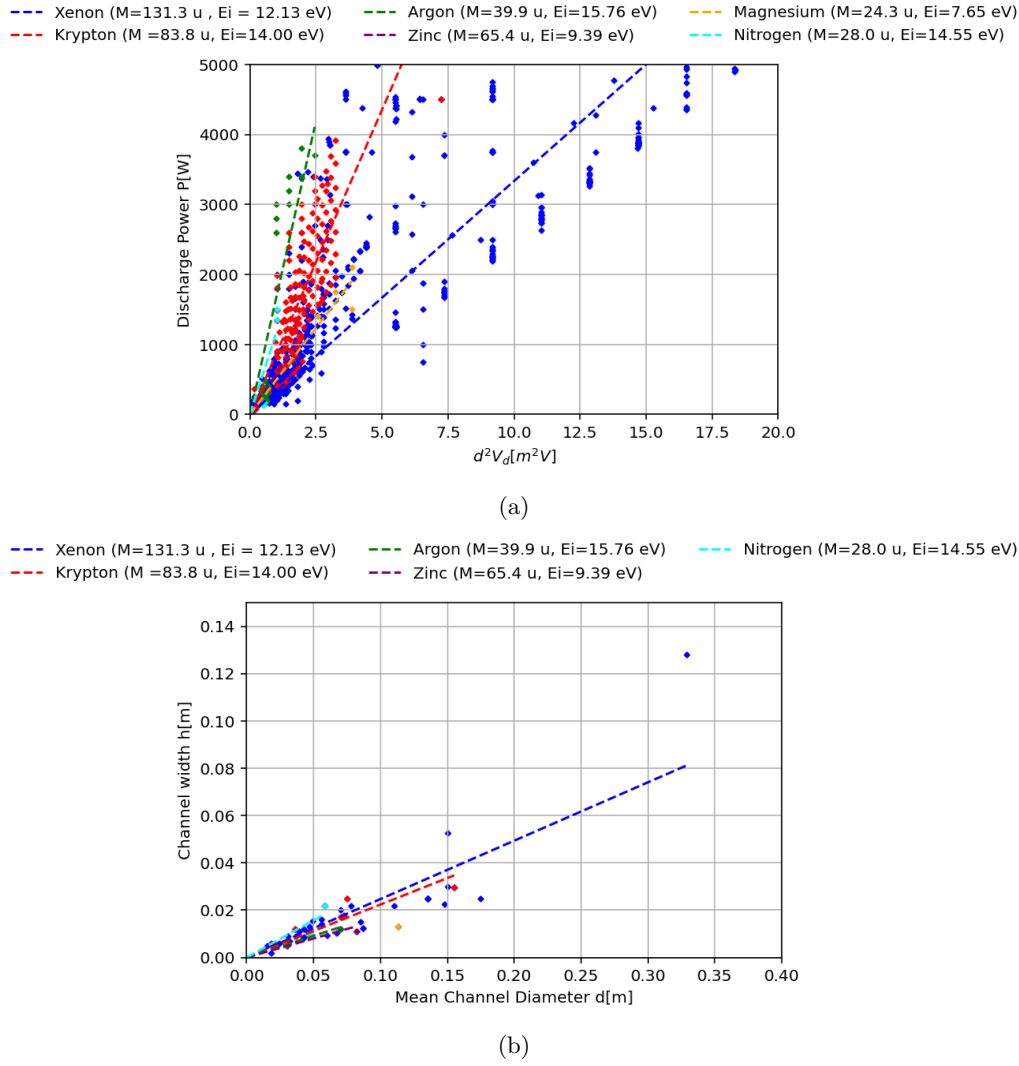


Figure 1.1: a) The discharge power scaling relationship. b) The channel width scaling relationship

Table 1.2: The scaling coefficients and channel cross-sectional dimensions for different databases and propellants.

	C_P	C_{hd}	d (mm)	h (mm)
SUEP, Zinc	585.3	0.1606	65.36	10.50
SUEP, Xenon	333.4	0.2468	86.59	21.37
SUEP, Krypton	870.4	0.2237	53.59	11.99
Lee et al., Xenon (2)	633.0	0.242	62.84	15.21

The scaling coefficients for zinc: C_P and C_{hd} allow an estimation for the dimensions of the Zinc Hall Effect Thruster, and from this, a channel diameter of 65.36mm and a channel width of 10.50mm are estimated. Comparing this to existing Hall Effect Thruster data, a reference for typical scaling coefficients across different propellants is obtained, allowing for trends to be examined which suggests that these coefficients can be linked to other variables such as mass and ionisation energy.

1.2.2 Ionisation and Channel Length

From Munro (3) there are three criteria for minimum channel length:

- It must be significantly greater than the Larmor radius of the electrons (r_e)
- It must be greater than the Debye sheaf (L_{sheaf})
- It must be significantly greater than the free ionisation path (λ_i), the Melikov-Morozov criterion

These are computed using the following assumptions made by Munro (3):

- $\approx 1.2 \times 10^{19} m^{-3}$
- $n_e \approx 1.2 \times 10^{18} m^{-3}$
- $24T_e \approx 200V_d$
- $L_{sheaf} \approx 5 - 10$

The Larmor radius of an electron (4) is calculated from Equation (1.5) (4) and displayed graphically in Figure 1.2. It is plotted for values of 100, 200 and 300 Guass to demonstrate a range of operating conditions. Clearly for operating at lower Guass, the Larmor radius, and so channel length, would need to be greater.

$$r_e = \frac{\sqrt{2m_e}}{|e|B} \quad (1.5)$$

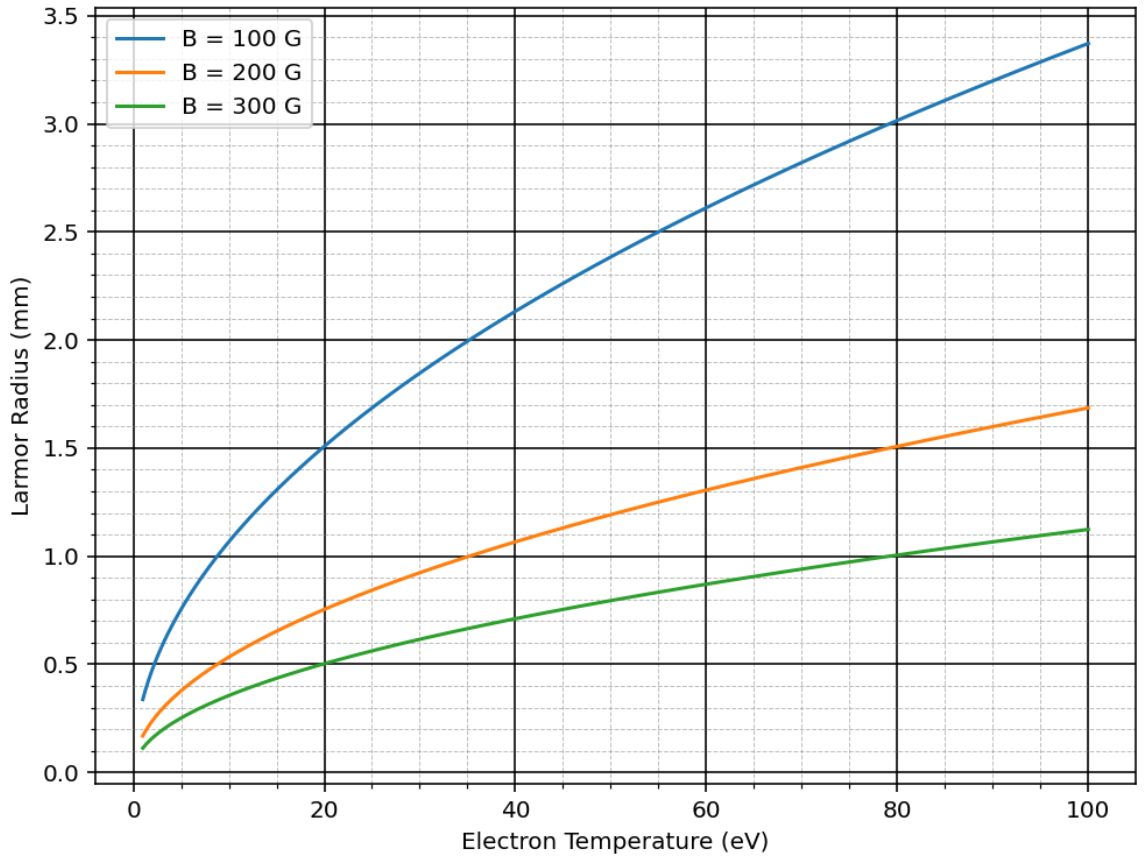


Figure 1.2: Larmor Radius

Hence for operating at 200V, the Larmor Radius - for the maximum B - is approximately 1.7mm. The minimum channel length for this criteria must so be 1.7mm.

Debye length is calculated from Equation (1.6) (4) and shown graphically in Figure 1.3.

$$\lambda_D = \sqrt{\frac{\epsilon_0}{n_e e^2}} \quad (1.6)$$

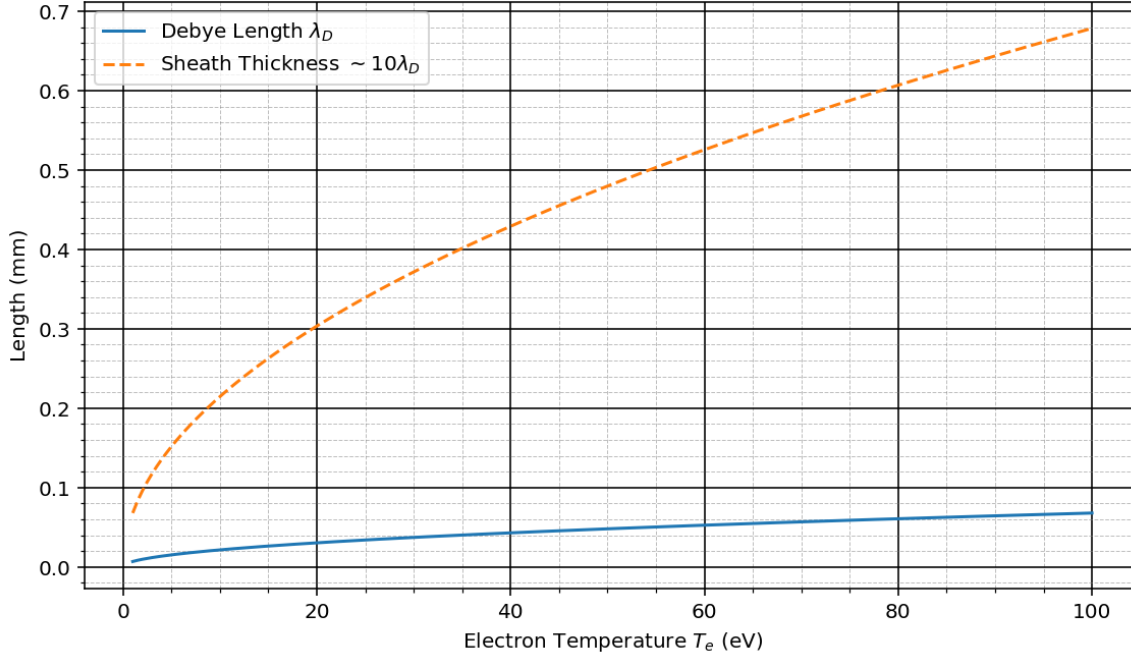


Figure 1.3: Debye Radius and Sheath as a function of electron temperature

Hence for operating at 200V, the Debye sheath is approximately 0.34mm. This is approximately 10x the Debye length at this voltage. The minimum channel length for this criteria must be 0.34mm.

To begin characterising λ_i , the ionisation rate coefficient $\langle \sigma_i v_e \rangle$ must first be found (4), using Equation 1.7 the numerically integrated and graphical results are shown in Figure 1.5.

$$\langle \sigma_i v_e \rangle = \int_{E_{th}}^{E_{max}} \sigma_i(E) v_e(E) f(E, T_e) dE \quad (1.7)$$

The cross section of several elements, $\sigma_i(E)$, is obtained from a dataset (5) and shown in Figure 1.4. Whereas $\nu_e(E)$, Equation (1.8) and $f(E, T_e)$, Equation (1.9), at certain electron temperatures in eV, are numerically integrated (4). The Maxwellian distribution is normalised over the dataset energy range to give a more accurate value from the limited dataset.

$$v_e(E) = \sqrt{\frac{2E}{m_e}} \quad (1.8)$$

$$f(E, T_e) = \frac{\sqrt{E} \exp\left(-\frac{E}{T_e}\right)}{\int_{E_{th}}^{E_{max}} \sqrt{E} \exp\left(-\frac{E}{T_e}\right) dE} \quad (1.9)$$

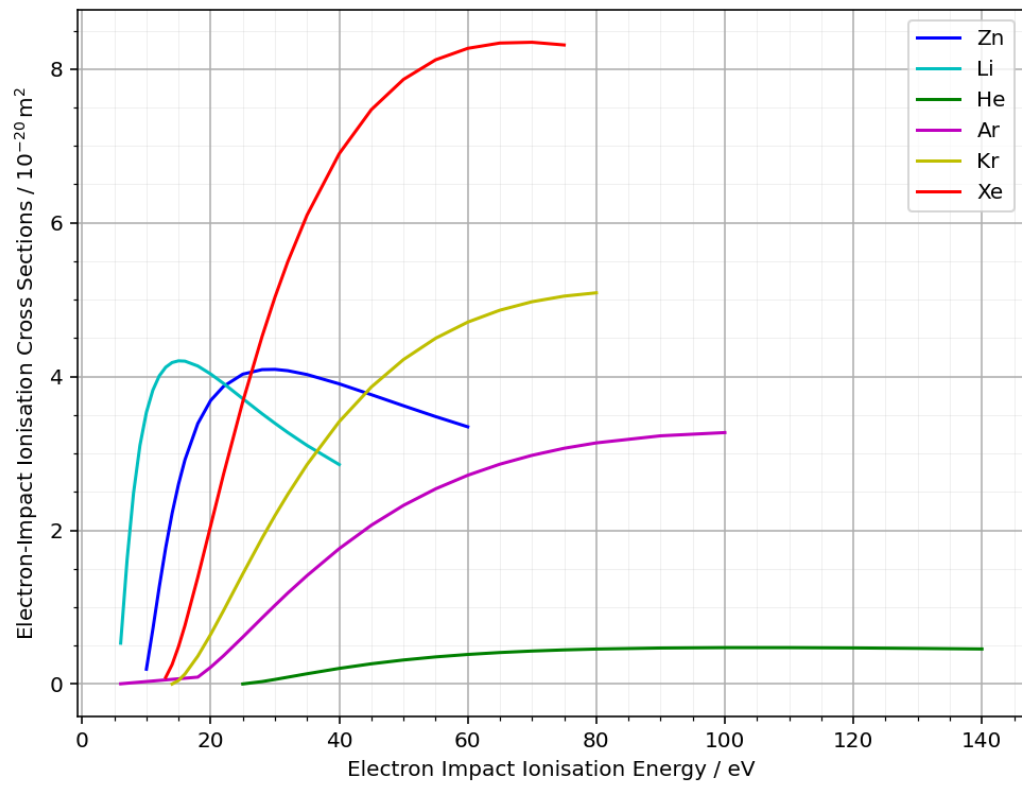


Figure 1.4: Electron-impact ionisation cross sections from incident electron energy at ground state

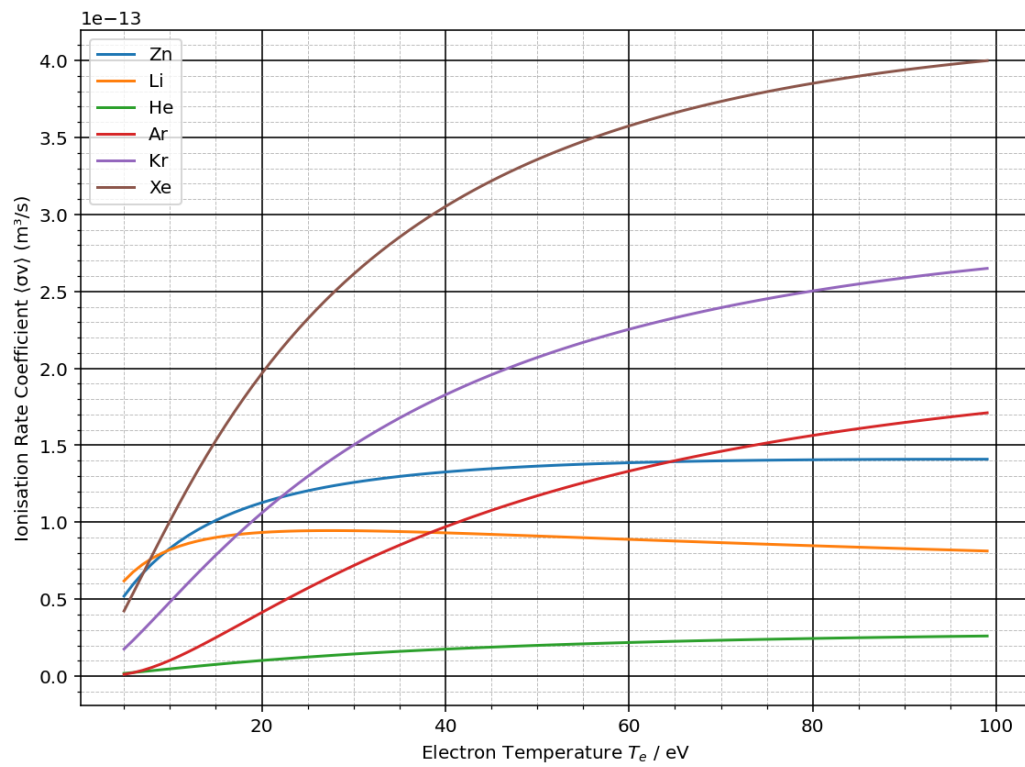


Figure 1.5: Electron-impact ionisation rate against electron temperature

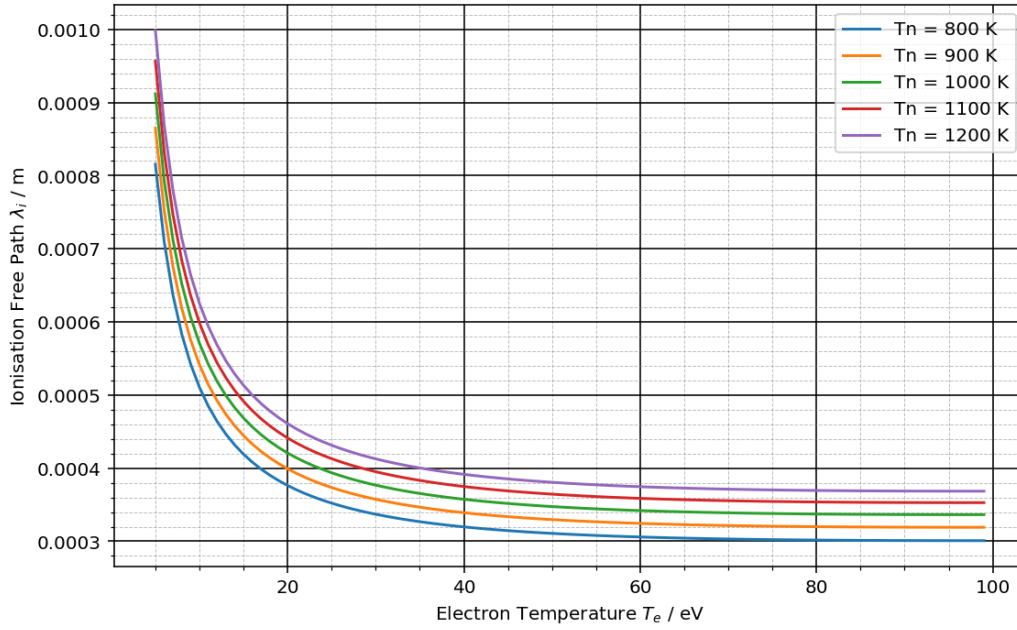


Figure 1.6: Ionisation Free Path

The Equation (1.10) (3) is used to estimate ionisation free path in at different neutral temperatures in Figure 1.6. High neutral gas temperatures are chosen to represent the higher temperatures due it being sublimated zinc.

$$\lambda_i = \frac{V_n}{V_i} = \frac{V_n}{n_n \langle \sigma_i v_e \rangle} \quad (1.10)$$

Where the typical speed of a neutral particle (4) is :

$$V_n = \sqrt{\frac{8k_B T_n}{\pi m_n}} \quad (1.11)$$

As the criterion states $L_c \gg \lambda_i$, at 1200K and 200V, the minimum channel length must be at least roughly 35mm, if it is assumed $L_c \approx 1000$.

As the free ionisation path demands the greatest minimum channel length of 35mm this must be the overall minimum channel length. As such, **the channel length is chosen to be 40mm** to satisfy this and any error.

1.2.3 Channel Material

When researching different properties of channel materials, thermal properties were given more attention than mechanical properties because the channel experiences extreme thermal cycling; they typically fail from thermal erosion. Data for these different properties have been compiled into tables D.1, D.2, D.3, and D.4.

High electrical resistivity electrically insulates the channel walls to suppress unintended short circuits, keeping the anode's positive charge. This helps to maintain the electric field structure and create a localised plasma at the channel exit. However, only a minimum insulation threshold has to be met. Therefore, electrical resistivity only has a weighting of 0.05. On the other hand, thermal conductivity should be as high as possible. It helps to efficiently dissipate heat and manage high

plasma loads. The heat spreading reduces the temperature gradient close to the plasma and inhibits high thermal-stress concentration hot spots. Because thermal conductivity has a significant effect on the channel temperature, erosion rates, and overall thruster lifetime, it has a high weighting of 0.20. A low coefficient of thermal expansion also prevents structural failure caused by cracking. Less thermal expansion also ensures channel dimensional stability which is integral for consistent discharge current and thruster performance. As a result, the coefficient of thermal expansion was set to 0.10.

The specific heat capacity governs how quickly the material heats up during thermal cycling; this delay buffers against rapid temperature gradients, reducing thermal stress. Though it needs to be considered, it does not strongly impact the erosion rate and temperature so it only scores 0.05. By contrast, a high maximum operating temperature is essential to maintain structural integrity without softening or decomposing at high temperatures. Importantly, it can withstand higher wall temperatures so operation life is longer and more efficient. It is then evaluated to have a weighting of 0.15. Though density does not have any direct effect on the thruster performance, it does affect thermal diffusivity: how quickly the channel responds to heating. Thus, it scores 0.05. More significantly, because material costs are limited to the project's budget, it is given the highest weighting of 0.25. Similarly, the ease of machining is pivotal as the thruster channel will be manufactured using the university's tools to save expenditure, earning it a 0.15 score.

1.2.4 Decision Matrix

To decide on the most optimal channel material, a decision matrix will be calculated. To first do that, normalized values for all the data are solved for in tables D.6, D.7, D.8, and D.9. For every column where higher values are optimal, the standard Min-Max normalizing formula below has been used.

$$\frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1.12)$$

However, for columns preferring lower values, the standard inverted Min-Max normalizing formula below has been used.

$$\frac{x_{\max} - x}{x_{\max} - x_{\min}} \quad (1.13)$$

As observed in tables D.10, D.11, D.12, and D.13, these normalised values are then multiplied by their respective weights. Next, each material gets a composite score using the summation of these weighted normalized values. Overall, Graphite Grade POCO AXF-5Q is quantified to be the ideal hall effect thruster channel material.

Table 1.3: Channel material properties composite scores

Material name	Composite score
Graphite Grade POCO AXF-5Q	0.7672
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.6140
Alumnium Nitride Ceramic Shapal Hi-M Soft	0.5563
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.5465
Boron Nitride-Silica Composite (BN-SiO ₂)Grade M	0.5109
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.4450
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.4166

1.3 Magnetic Circuit

1.3.1 Selection of Magnetic Circuit Type

The magnetic circuit for a hall thruster in its most basic form should produce a radial field close to the exit plane, which will serve to "trap" electrons. In a hall thruster, there are two types of magnetic field configurations, shielded and non-shielded, along with two fundamentally different ways to produce the magnetic circuit, either using electromagnets or permanent magnets.

Initial requirements for this thruster stipulate that magnetic shielding will be attempted, which involves curving the magnetic field lines around the discharge channel, so that charged particles experience a deflective force that reduces the chance of wall collisions. This results in a lower wall temperature and reduced potential on the walls, which is illustrated by Figure 1.7.

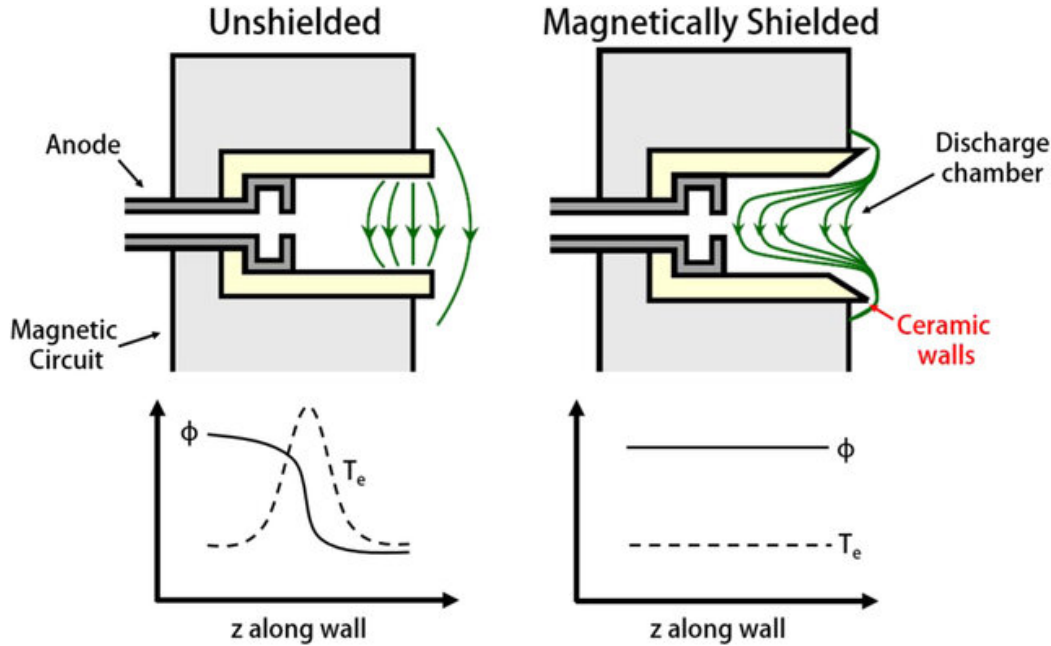


Figure 1.7: Illustration showing the difference in magnetic field line structure for a non shielded thruster (left) and a shielded one (right) (6).

To produce the desired magnetic field configuration, both permanent magnets and electromagnets were considered. Below is a decision matrix that determines the most appropriate option for our

application.

Table 1.4: Decision matrix for type of magnetic circuit to use, numbers through from 1 – 5 are assigned with 1 being the lowest score and 5 being the highest.

Variable	Weighting	PM	EM	PM Final	EM Final
Power	3	5	1	15	3
Adaptability	5	1	5	5	25
Complexity	2	5	3	10	6
Field strength	5	3	5	15	25
Manufacturability	4	4	3	16	12
Total:	-	-	-	61	71

Table 1.4 shows that the circuit will be an electromagnetic circuit for the main reasons of being able to achieve a wide range of field strengths and being able to achieve much stronger magnetic fields. Using electromagnets also makes the startup of the thruster more reliable.

1.3.2 Iterative Magnetic Field Design

1.3.3 Mesh Independency Study

1.4 Anode Design and Gas Distributor

1.5 Virtual Integration

1.6 Thermal Simulation

Chapter 2

Cathode

[Numbers for Richardson Dushman are wrong. Rewrite the temperature and dimensions][New Values : $T_c = 1730K$]

2.1 Requirements Definition

The purpose of the cathode is to match the current of positive charge being emitted by the thruster with an equal current flow. Without it, the thruster will become negatively charged over time and it will reduce the efficiency of the thruster as the ions will be attracted back to the thruster. It is also necessary to produce the virtual cathode effect that the main thruster relies on.

The first thing to define is the operating current range which can be determined by the nominal power requirement of the thruster and the expected voltage requirement of the thruster. Since the cathode current needs to match the thruster current, cathode current can be calculated to range between $0.83 - 10A$.

2.2 Plasma and Emission Physics

The estimated current density of the insert can be determined using the material work functions and the Richardson-Dushman equation.

$$J = AT^2 e^{\frac{-e\phi}{kT}} \quad (2.1)$$

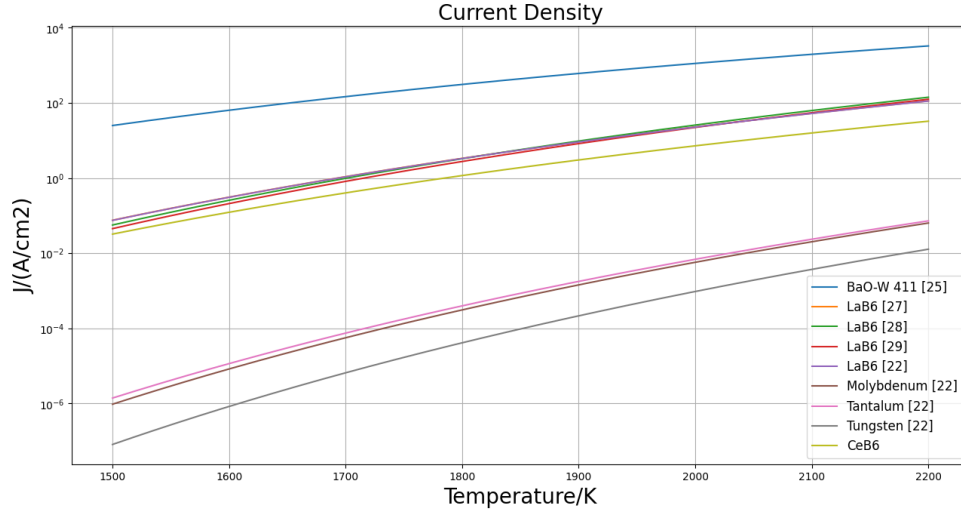
This equation describes the output current density of a material with respect to the temperature and its material properties. J is the output current density, A is a material constant, e is the elementary charge of an electron, T is the temperature of the material, ϕ is the work function of the material (which is also a function of the temperature), and k is the Boltzmann Constant (1.38×10^{-23}).

Using the work function temperature relations described in the table below.

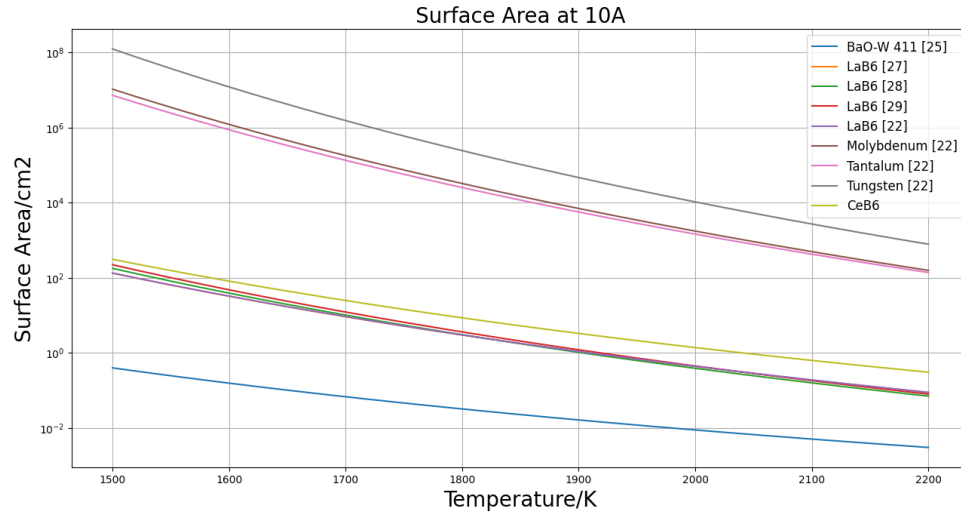
The following plots can be derived from the calculated work functions and the expected current range.

	A	D	ϕ
<i>BaO – W</i> 411	120	-	$1.67 + 2.82 \times 10^{-4}T$
<i>LaB₆</i>	-	29	2.66
<i>LaB₆</i>	-	110	2.87
<i>LaB₆</i>	120	-	2.91
<i>LaB₆</i>	120	-	$2.66 + 1.23 \times 10^{-4}T$
Molybdenum	-	55	4.2
Tantalum	-	37	4.1
Tungsten	-	70	4.55

Table 2.1: Work Function



(a) Current Density



(b) Surface Area

We can also start to work out the expected lifetime that the cathode needs to be active for using the lifetime of the fuel source. This helps to give us a target lifetime to start trying to model the insert thickness.

To start we looked at models of material degradation for the better performing materials and started defining the insert dimensions. To start we decided on an appropriate insert length ($5mm$) and used the calculated surface area to calculate the required inner diameter. We then used some

experimental data of LaB_6 evaporation rate to calculate the total required mass to last the required minimum lifetime of 100hrs.

We decided to remove $BaO - W$ from consideration due to its inability to tolerate impurities in the feed gas, toxicity issues and manufacturing difficulties. LaB_6 has shown significantly greater tolerance to gas impurities.

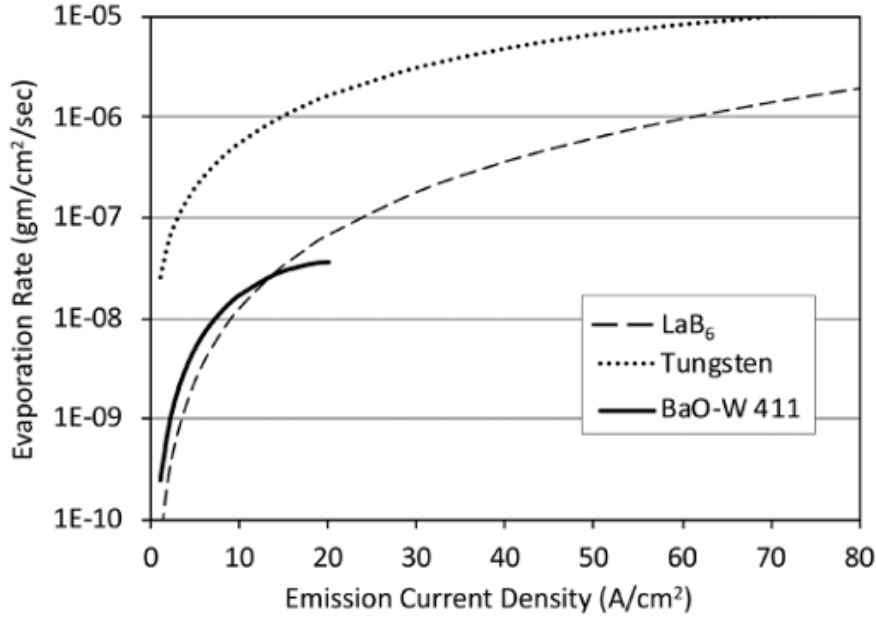


Figure 2.2: Evaporation Rate of Insert Materials

We can use the figure above to estimate the required mass of the insert and we can then use the required inner surface area calculated earlier with the decided insert length to calculate the inner and outer diameter of the insert.

That lead to the calculation that at an optimal temperature of 1900K, the thickness of the insert would be approximately 0.004mm, for the sake of manufacturing it was decided to use a thickness of 1mm. This lead to an estimated lifetime of $9.84 \times 10^4 hrs$.

The current dimensions of the insert are:

- Inner Diameter = 5mm
- Outer Diameter = 7mm
- Length = 8mm

Current Orifice radius estimate = 0.4mm

2.3 Neutral Gas Flow and Orifice

To determine the expected state of the plasma at all points of the cathode, a 0-D model was implemented.

The current model is largely inspired from the "Siegfried and Wilbur" model discussed in (7). This model first involves calculating the pressure at the orifice using isentropic flow estimations. This leads to inaccuracies as it doesn't include the plasma pressure side of things.

The next step is to get an estimate for the electron number density using empirical values for the ionisation fraction at the orifice (α), a plot is provided.

Then using a "two temperature Saha Model" equation that looks like

$$\frac{\alpha^{1+\frac{T_g}{T_e}}}{(1-\alpha)^{\frac{T_g}{T_e}}(1+\alpha\frac{T_e}{T_g})} = \frac{1}{P} \frac{(2\pi m)^{3/2}}{h^3} e^{5/2} T_g T_e^{3/2} \left(\frac{\sum_i}{\sum_0} \right)^{\frac{T_g}{T_e}} \exp\left(-\frac{\epsilon_i}{T_{eV}}\right) \quad (2.2)$$

This equation is used to solve for the electron temperature.

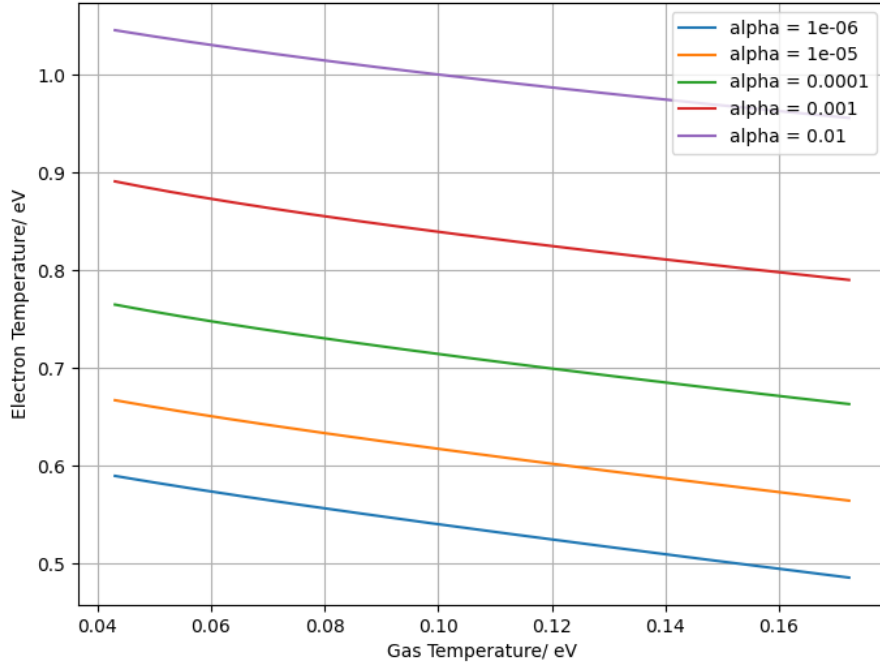


Figure 2.3: Electron Temperature alpha range

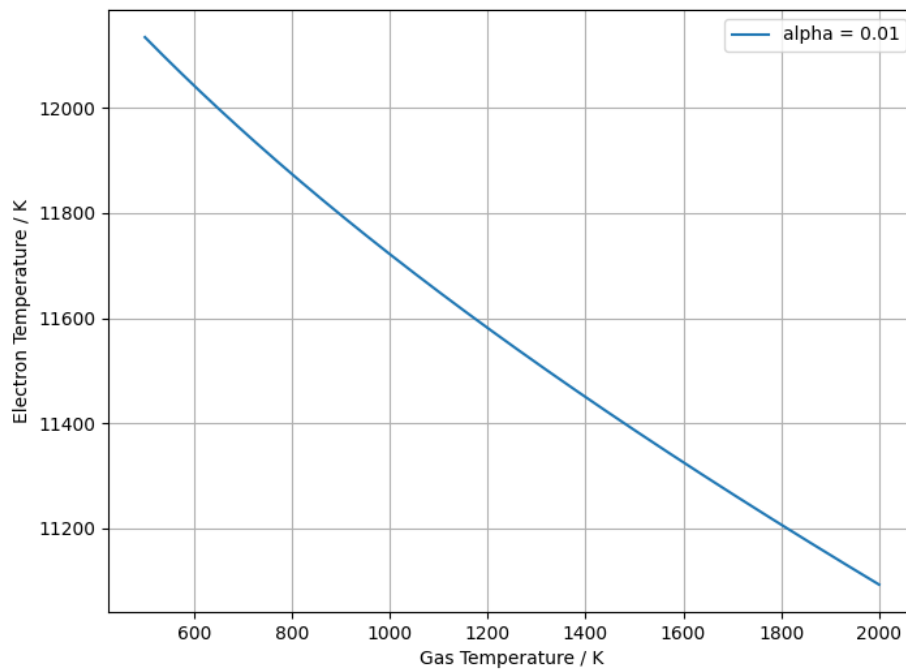


Figure 2.4: Electron Temperature fixed α

2.4 Ignition and Breakdown

2.5 Geometry and Component Sizing

2.6 Thermal Management

2.7 Lifetime and Degradation

2.8 Integration and Interfaces

Chapter 3

Propellant Delivery

3.1 Requirements Definition

The propellant storage and delivery system (PSDS) is designed to sublime the solid zinc fuel at a specific mass flow rate to feed to the cathode and thruster subsystems. The design parameters of the PSDS system are shown in Table 3.1. The parameters are based on calculated values in Sections 1.1, 2.1 and 3.2.

Table 3.1: The input design parameters to be used in scaling the size of the PSDS.

Design Parameter		Value
Power Consumption		50W
Vapour Pressure Range		$(0.01 - 2.5)Pa$
Temperature Range		$(280 - 420)^{\circ}C$
Optimum Propellant		Zinc
Mass Flow Rate	$2.22 - 3.55mg/s$	or $45.71 - 72.97sccm$

These parameters are designed to maintain a time-dependent stable and controllable state to satisfy the mass flow rate requirements of the Zinc cathode and thruster. The system must be chemically inert throughout its lifetime and designed to be practically manufacturable and accessible post-assembly. The controllability of the system can be defined as the ability to provide a required mass flow rate at any given point in time, within the operating envelopes. The stability of the system can be defined as the ability of the PSDS to maintain a time-invariant mass flow rate across the operating time alongside the ability to passively and/or actively resist runaway evaporation and temperatures causing runaway mass flow rates.

3.2 Vapour Pressure and Operating Envelope Definitions

3.2.1 Vapour Pressure and Mass flow rate literature verification

The vapour pressure of an element is an intrinsic property that allows one to know the flux of particles sublimating from the surface of a solid/liquid phase at any moment. The vapour pressure

of an element is only dependent upon the temperature of the specimen. The vapour pressures of any given element at any given temperature and phase were empirically derived by C. Alcock. The relationship derived is seen below in Equation 3.1.

$$\log(P_v) = 5.006 + A + B + T^{-1} + C \log(T) + DT^{-3} \quad (3.1)$$

Where P_v is the vapour pressure, T is the temperature and A , B , C and D are empirical coefficients for Zinc in solid form, taken from (8).

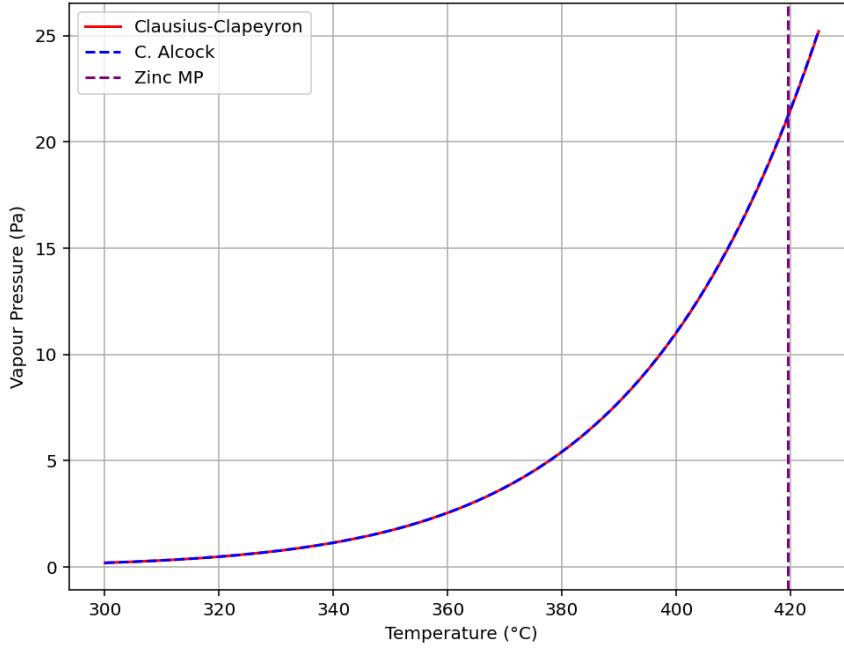


Figure 3.1: Shows the vapour pressure against temperature range

The corresponding figure illustrating the V_p vs T relationship is Figure 3.1. Illustrated on the graph is the melting temperature of Zinc (T_{m_z}) - 419.5 deg C. Since the propellant phase is taken as a solid, the temperature of the specimen used must remain below T_{m_z} as the empirical coefficients being used in the C. Alcock relation (Equation 3.1) are for the solid phase. To verify the relationship, it was compared to the Clausius-Clapeyron relationship below (Equation 3.2), derived from the laws of thermodynamics, as seen in Figure 3.1.

$$P_v = P_{ref} e^{-\frac{\Delta H_s}{R} \left(\frac{1}{T_1} - \frac{1}{T_{ref}} \right)} \quad (3.2)$$

Where ΔH_s is the latent heat of sublimation for Zinc and P_{ref} and T_{ref} are reference pressure and temperature values. The reference Temperature value was taken as 603 K and the reference pressure derived from the C. Alcock relationship using the reference temperature. Note that the reference temperature is required to be taken close to the temperature range of interest to obtain the best fit due to the simplification of linearising (negating temperature dependence) the enthalpy of sublimation, $\Delta H_{sublimation}$. Please also note $\Delta H_{sublimation}$ is calculated from the B coefficient in the C.Alcock relation.

Through the Hertz-Knudsen equation (Equation 3.3) the mass flow rate of particles leaving the surface can then be found using the vapour pressure relationship just derived.

$$\dot{m} = \underbrace{\eta \frac{P_v(T_s)A}{\sqrt{\frac{2\pi k_B T_s}{M}}}}_{\text{Sublimation MFR}} - \underbrace{\eta \frac{P_c A}{\sqrt{\frac{2\pi k_B T_s}{M}}}}_{\text{Condensation MFR}} \quad (3.3)$$

Where $\dot{m}$ is the mass flow rate, T_s is the temperature of the solid Zinc, $P_c = 0.01 Pa$ is the pressure of the chamber, k_B is the Boltzmann constant, M is the molecular mass of Zinc and A is the exposed area of the solid surface. The vapour pressure (P_v) was found via Equation 3.1. Below (Figure 3.2) is a graph illustrating the impact of varying the efficiency whilst keeping all other parameters constant.

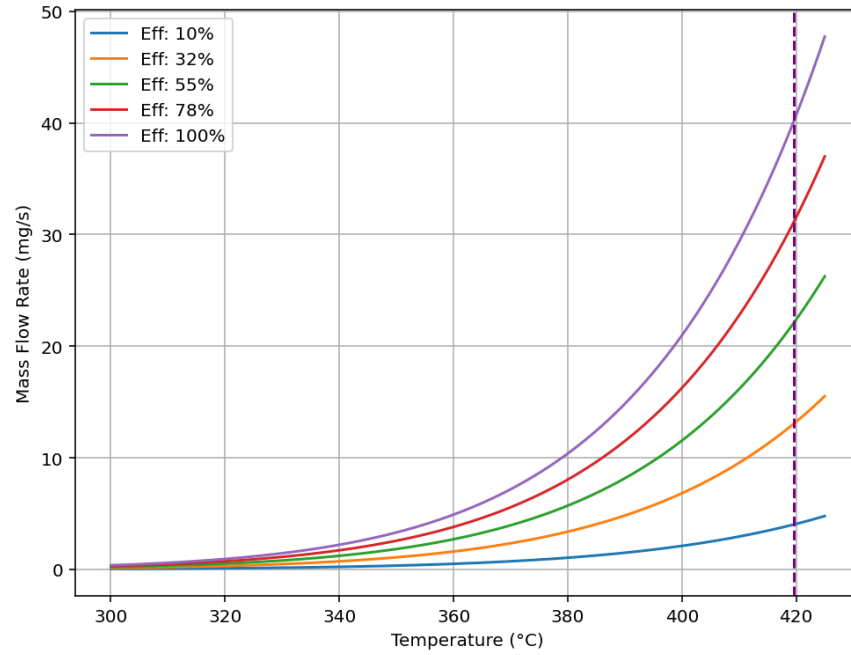


Figure 3.2: Mass Flow Rate vs temperature illustrating effect of sublimation efficiency for a surface area of 1134 mm^2

Based on previous literature, a comparison study between the mass flow rates the graphs produced illustrate and those obtained in the literature experiments for a hollow cylindrical Zinc slug of dimensions $\text{Ø}16 \times \text{Ø}6 \times 19 \text{ mm}$ utilising a cartridge heater. The initial temperatures required to obtain mass flow rates from a cylindrical propellant slug were then obtained and are illustrated in Figure 3.3. The surface area of sublimation was taken as the combined area of the inner and outer surfaces of the cylinder, totalling to 1134 mm^2 . Note the mass flow rate is also dependent upon the surface area of the specimen. Therefore, the graphs looked at the effects of adjusting the surface area by increasing/decreasing it by 50%. Further analysis of this behaviour between area, temperature and mass flow rate is shown in Figure 3.4, which shows that as surface area and temperature increase, there is an increase in mass flow rate, as expected.

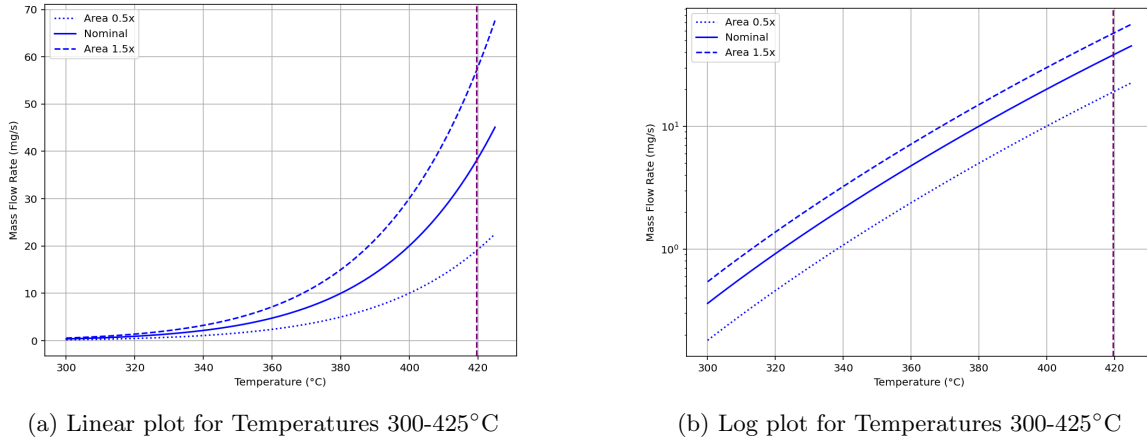


Figure 3.3: Shows the Mass Flow Rate (MFR) against temperature plots, accounting for changes in area.

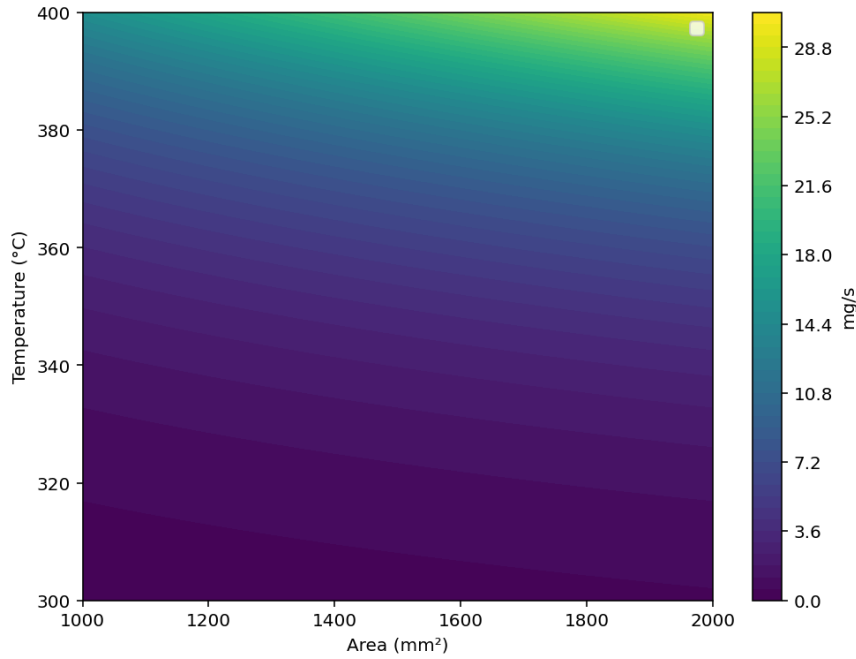


Figure 3.4: Shows a contour plot of the Temperature vs Area relationship to MFR of Zinc.

In this case of comparison, since the exposed area of sublimation is taken to be the 'side' area of the cylinder, it is evident the surface area will change over time, as the exposed material sublimates away. To maintain a constant mass flow rate, this means the temperature of the specimen's sublimation surface will need to be changed over time. Hence, the Hertz-Knudsen, density and C. Alcock relations were arranged through a set of differential equations to observe the required change in temperature over time to maintain a constant mass flow rate. The mass flow rate coupled with the known density and volume values and relations of the propellant slug were used to obtain the rate of change of the radius over time, $\frac{dr}{dt}$, and hence the changing exposed area over time, $\frac{dA}{dt}$. The Hertz-Knudsen coupled with the C. Alcock relation was used to obtain the change in area, $\frac{dA}{dt}$. Hence the equation below could be formed and solved through time-stepping to obtain the changing temperature requirements over time to maintain a constant mass flow rate.

$$\frac{dT}{dt} = \frac{dA}{dt} \frac{dT}{dA} \quad (3.4)$$

Note in the formation of these results and graphs, the temperature gradient across the specimen and sublimation efficiency are yet to be accounted for. A further simplification was made in that the inner radius is not experiencing sublimation and maintains optimum heater contact throughout the total sublimation cycle. Furthermore, the top and bottom circular surfaces are taken as non-sublimating surfaces. Both these assumptions are in practise not true and do need to be accounted for. However, the general observation can be made in that the temperature of the heater is required to be increased over time as the sublimation area decreases, which was an expected outcome.

3.3 Compatible Material Analysis

3.3.1 Material Criteria

The PSDS features a separate reservoir for solid zinc storage, which will be heated to temperatures capable of sublimating the zinc. From this, the gaseous zinc will be fed through a feed line, connecting the reservoir to the thruster and cathode. The attributes of the reservoir material to consider are as follows:

- i. **Price:** A primary concern is cost, and as such a balance must be met between the price and performance of the material. This also stems from the ability to source the material and machine it to design specifications.
- ii. **Thermal Conductivity:** The material must balance between acting as a thermal insulator and possessing the ability to conduct heat to allow sublimation of the solid zinc fuel to occur. Alternatively, an insulating layer could be applied around the reservoir walls. As its temperature increases, its thermal conductivity decreases.
- iii. **Specific Heat Capacity:** Both high and low specific heat capacity values must be considered. A low specific heat capacity would allow for rapid heating of the reservoir walls; however, a high specific heat capacity would handle the high temperature spikes produced by zinc sublimation, which in turn maintains a stable environment, contributing to performance and lifetime.
- iv. **Melting Point:** To contain the zinc gaseous vapour, the material must have the ability to handle the high temperatures required to sublimate the solid zinc propellant. Any indication of the reservoir material approaching its melting point will compromise its structural integrity and that of any adapters, fixtures, or transducers attached to the reservoir body itself. Another concern would be contamination of the zinc propellant by outgassing of the reservoir material, as the reservoir material will be subjected to both a low pressure and high temperature environment.
- v. **Thermal Expansion Ratio:** When heated during operation and cooled when finished, the reservoir material will expand and contract, respectively. Depending on the reservoir material,

internal cracks or weakening of the grain structure may occur as a result of this cyclic process, leading to structural fatigue and warping.

- vi. **Electrical Conductivity and Resistivity:** As the reservoir will be heated by electrical means, the material must be able to effectively conduct electrical current. For most metals, as its temperature increases, its electrical conductivity decreases; meanwhile, its electrical resistivity increases.

3.3.2 Materials Analysis

Selection of a suitable material for the reservoir was complete using Granta EduPack 2025. A full comparison of a wide range of materials related to the attributes listed in Section 3.3.1 and constraints below is laid out as follows:

- Minimum melting point temperature: $\approx 850^{\circ}\text{C}$
- Maximum operational temperature: 420°C

These constraints are derived from the initial requirements defined in Section 3.1, shown in Table 3.1. The maximum operational temperature is already outlined in Table 3.1, while the Minimum melting point temperature was calculated by taking the maximum temperature range, and applying a safety factor of 2.

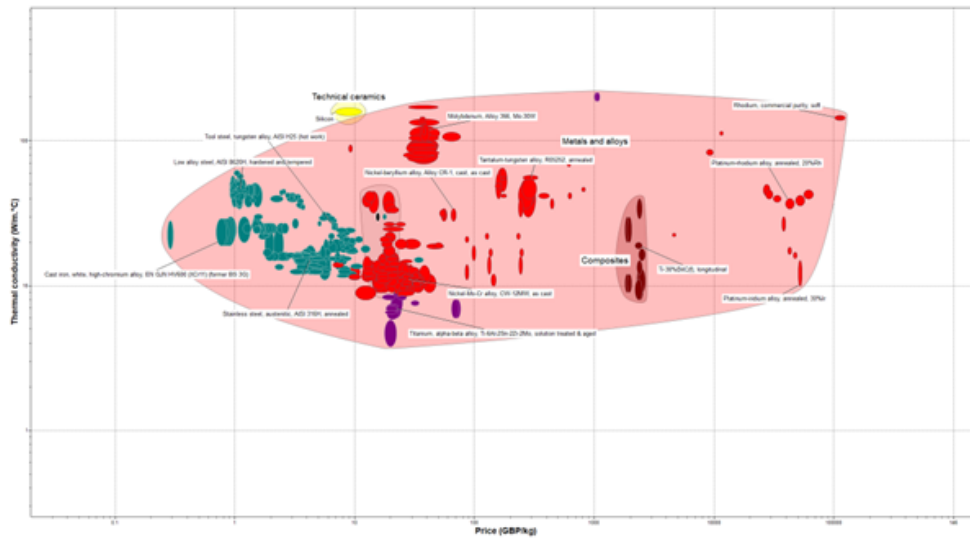


Figure 3.5: Thermal conductivity against price.

Figure 3.5 shows the thermal conductivity against price for a range of metals. Ferrous metals and alloys such as stainless steels (e.g., 304, 316), low alloy steels, and tool steels offer moderate thermal conductivity while remaining at a relatively low price compared to groupings such as non-ferrous metals (e.g., nickel molybdenum alloys), refractory metals (e.g., commercially pure molybdenum), rare earth metals, and precious metal alloys. Technical ceramics (e.g., silicon) offer the highest thermal conductivity, while composites possess similar thermal conductivity properties comparable to ferrous metals while also being more expensive per kilogram.

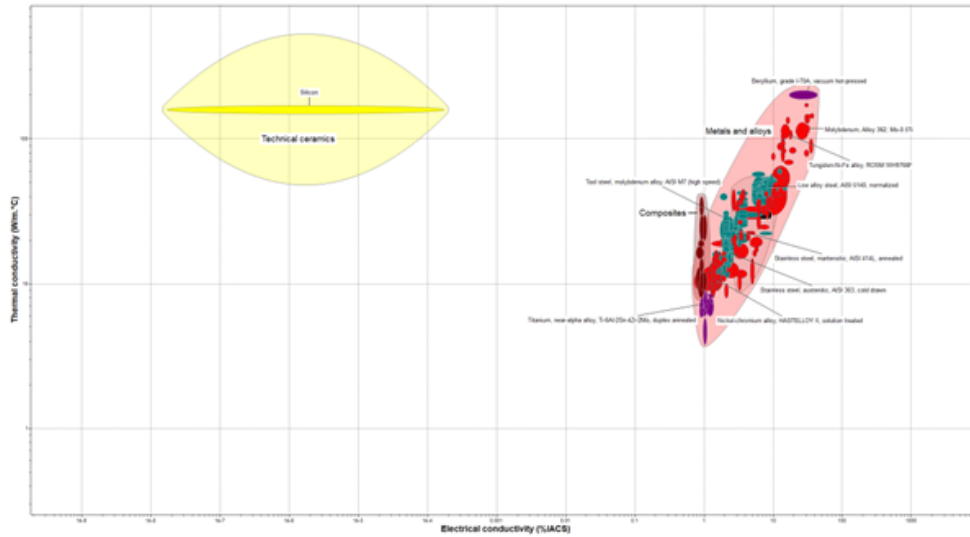


Figure 3.8: Melting point against thermal conductivity.

Figure 3.8 compares thermal conductivity against electrical conductivity. Silicon shows the least thermal conductivity, while metals and alloys show the highest electrical conductivity. However, the thermal conductivity of these metals and alloys varies, with composites resting on the lower end and materials like molybdenum alloys located at the higher end. As seen in previous trends, ferrous metals sit within a moderate range, indicating a balance of attributes.

3.3.3 Decision Matrix

Table 3.2 shows a weighted decision matrix detailing a holistic comparative view of material representatives.

Table 3.2: Weighted materials decision matrix (Scale: 1-5) of representative materials.

Criteria	Weight	SS 304	SS 316	Molybdenum
Price	0.30	2	3	4
Thermal Cond.	0.15	3	2	5
Specific Heat Capacity	0.15	3	3	2
Electrical Cond.	0.10	-	-	-
Electrical Resistivity	0.10	-	-	-
Machinability	0.10	4	3	1
Score	-	-	-	-

3.4 Coatings Material Analysis

3.4.1 Coating Criteria

Prior attempts at a PSDS have mentioned the desire for the use of coatings to help inhibit Zinc deposition, while assisting the systems functionality and longevity (9). Unfortunately, limited research exists about coatings for a PSDS, with Zinc gas at high temperatures. However, some analysis of

coatings that is known not to react with Zinc, that function at high temperatures can be done to assist in picking a desirable candidate to test with and evaluate later on. First some key criteria that will influence this decision include:

- i. **Price:** Coatings will range in price from the materials used and application methods. Therefore, a balance between price and performance needs to be taken into account as money is a valuable resource in this project.
- ii. **Reactivity:** The coating should not react with Zinc, particularly at high temperatures. This ensures that the propellant is used up for its desired purpose and no unexpected outcomes occur. Additionally, it would be desirable if it inhibited the deposition of Zinc.
- iii. **Thermal Properties:** Ideally, the coating will reduce the thermal gradients experienced in the internal surfaces. This will ensure longevity and efficient functioning of the PSDS.
- iv. **Porosity & Bonding:** This is a measure of quality, where a highly porous coating will have more of the surface under, exposed resulting in limited effectiveness. Bonding refers to the ability for the coating to stick to the material which it is applied too, which can further measure its quality. Therefore a coating with low porosity and based on Section 3.3, a coating that bonds well to stainless steels is key.
- v. **Durability:** If Zinc does deposit itself, it should be able to withstand the process of removing the solid Zinc and any scraping. Preferably it would inhibit this

3.4.2 Coating Analysis

The potential for a coating that will provide protection against the harsh thermal environment that will be seen in the PSDS, while also inhibiting zinc deposition, is mentioned in sources as an ideal and useful insight to gain (9). Ideal properties of a coating in this application include the ability to reduce thermal gradients, non-reactive to zinc, low cost and high quality (low porosity). There is a lack literature sources that delve into coatings applicable to an environment which possesses high temperature zinc gas. Despite this it would be useful to test a coating to see its effects. Therefore, analysis of different coating and their properties in temperature ranges expected for molten zinc have been analysed to give viable candidates.

3.4.3 Decision Matrix

Table 3.3: Decision matrix for PSDS coating selection, where the total is the weighting times the score given.

Variable	Weighting	Alumina Oxide	Electromagnet	$PM \times$ <i>weighting</i>	$EM \times$ <i>weighting</i>
Price	5	-	-	-	-
Reactivity	5	-	-	-	-
Thermal prop.	4	-	-	-	-
Porosity	3	-	-	-	-
Bonding	3	-	-	-	-
Total:	-	-	-	61	71

3.5 Accessibility and modularity

To ensure the Propellant Storage and Delivery System (PSDS) is maintainable, the design is split into three modular zones: a reservoir, a transport line and a distributor. This allows for independent adjustments to zinc storage capacity or heater configurations without a full thruster rebuild. To prevent permanent blockages from zinc condensation, the propellant path avoids welds in favour of standard flanges or graphite gaskets to allow for disassembly and cleaning. A three-point temperature monitoring system—with sensors at the reservoir, transport line, and anode interface—is used to monitor the propellant state while minimising potential vacuum leak paths. The operating temperature range of 280 °C to 420 °C is required to maintain the vapour pressure necessary for Zinc sublimation while preventing re-solidification. Because the metal delivery pipes will physically expand at these temperatures, the design could include U-bends to act as springs, preventing expansion stresses from cracking the discharge channel. To ensure system longevity, materials like Aluminium or Stainless Steel should be avoided for any component in direct contact with Zinc vapour due to aggressive chemical reactions at high temperatures.

3.6 Zinc Storage and Feedstock

3.7 Zinc Sublimation Control

3.8 Flow Routing and Modularity

3.9 Contamination and Deposition Control

3.10 Thermal Integration

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Appendix A

Database

For this project, a large database of thrusters was curated. Below is a list of all the thrusters used. Data can be found publicly available online and the database itself can be found at this URL: [ENTER URL](#).

50–100 W Hall thruster	HEKT-100	PPS-1350-S
6kW IHI	HEPS-200	PPS-20k ML
A-3	HEPS-500	SHARK-600V
BHT-1000	HET-100	Silverlight
BHT-1500	HET-1350PM	SPT-20
BHT-1500-C	HET-1350PM V2	SPT-25
BHT-200	KHT-40	SPT-30
BHT-600	KHT-50	SPT-50
BHT-600-I	KM-32	SPT-50M
BHT-Bi-1500-V2	KM-37	SPT-70
BPT-2000	MaSMi60	SPT-100
CAM200	MaSMiDm	SPT-100-FLIGHT
CHT-Zn	Music-si	SPT-140
Ha-HET	NASA-173Mv1	SPT-200
H10	NASA-173Mv1 Anode 2	THT-VI
H6	NASA-173Mv2	TUD-H3-P
H9	NASA-457 HET	UAH-78AM
H9 MUSCLE	PPS-1350-E	UM/AFRL P5

Appendix B

Mesh Independancy Studies

Appendix C

Risk Assessments

Appendix D

Channel Material Tables

Table D.1: Channel material properties

Material name	Electrical resistivity ($\cdot\text{cm}$) at 25 degrees	Thermal Conductivity ($\text{W}/\text{m}\cdot\text{K}$)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	1.00E+14 (10)	14 (10)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	1.00E+14 (10)	29 (10)
Alumina (Al ₂ O ₃) CeramAlox 99.7	1.00E+14 (11)	33 (11)
Hexagonal Boron Nitride Composite (h-BN) Grade HP	1.00E+13 (10)	29 (10)
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	1.00E+14 (10)	130 (10)
Alumnum Nitride Ceramic Shapal Hi-M Soft	1.00E+15 (12)	92 (12)
Graphite Grade POCO AXF-5Q	1.47E-03 ()	95 ()

Table D.2: Channel material properties

Material name	Coefficient of thermal expansion ($\mu\text{m}/\text{m}\cdot\text{K}$)	Specific heat capacity ($\text{J}/\text{kg}\cdot\text{K}$)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.4 (10)	760 (10)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.1 (10)	770 (10)
Alumina (Al ₂ O ₃) CeramAlox 99.7	7.5 (11)	880 (13)
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.8 (10)	810 (10)
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	1.1 (10)	810 (10)
Alumnum Nitride Ceramic Shapal Hi-M Soft	5.0 (12)	480 (14)
Graphite Grade POCO AXF-5Q	7.9 ()	700 ()

Table D.3: Channel material properties

Material name	Maximum use temp inert (°C)	Density (kg/m ³)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	1000 (10)	2300 (10)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	1000 (10)	2100 (10)
Alumina (Al ₂ O ₃) CeramAlox 99.7	1700 (11)	3950 (15)
Hexagonal Boron Nitride Composite (h-BN) Grade HP	1150 (10)	2000 (10)
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	2000 (10)	1900 (10)
Aluminium Nitride Ceramic Shapal Hi-M Soft	1900 (12)	2880 (12)
Graphite Grade POCO AXF-5Q	2000 ()	1780 ()

Table D.4: Channel material properties

Material name	Material Cost (£/in ³)	Machining ease (scored 1-10 descending order)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	18.00	8.00
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	18.00	8.00
Alumina (Al ₂ O ₃) CeramAlox 99.7	1.76	1.00
Hexagonal Boron Nitride Composite (h-BN) Grade HP	85.72	6.00
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	150.00	6.00
Aluminium Nitride Ceramic Shapal Hi-M Soft	75.05	4.00
Graphite Grade POCO AXF-5Q	0.034	10.00

Table D.5: Channel material properties weighting

Material property	Weighting
Electrical resistivity	0.05
Thermal conductivity	0.20
Coefficient of thermal expansion	0.10
Specific heat capacity	0.05
Max operating temperature	0.15
Density	0.05
Material Cost (£/in ³)	0.25
Machining ease (scored 1-10 descending order)	0.15

Table D.6: Channel material properties normalized values

Material name	Electrical resistivity normalized values	Thermal Conductivity normalized values
Boron Nitride-Silica Composite (BN-SiO ₂)Grade M	0.1000	0.0000
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.1000	0.1293
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.1000	0.1638
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.0100	0.1293
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.1000	1.0000
Alumnum Nitride Ceramic Shapal Hi-M Soft	1.0000	0.6724
Graphite Grade POCO AXF-5Q	0.0000	0.6983

Table D.7: Channel material properties normalized values

Material name	Coefficient of thermal expansion normalized values	Specific heat capacity normalized values
Boron Nitride-Silica Composite (BN-SiO ₂)Grade M	0.9615	0.7000
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	1.0000	0.7250
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.0513	1.0000
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.9103	0.8250
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.8718	0.8250
Alumnum Nitride Ceramic Shapal Hi-M Soft	0.3718	0.0000
Graphite Grade POCO AXF-5Q	0.0000	0.5500

Table D.8: Channel material properties normalized values

Material name	Maximum use temp inert normalized values	Density normalized values
Boron Nitride-Silica Composite (BN-SiO ₂)Grade M	0.0000	0.7604
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.0000	0.8525
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.7000	0.0000
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.1500	0.8986
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	1.0000	0.9447
Alumnum Nitride Ceramic Shapal Hi-M Soft	0.9000	0.4931
Graphite Grade POCO AXF-5Q	1.0000	1.0000

Table D.9: Channel material properties normalized values

Material name	Material Cost (£/in ³)	Machining ease (scored 1-10 descending order)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.8802	0.7778
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.8802	0.7778
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.9885	0.0000
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.4286	0.5556
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.0000	0.5556
Alumnum Nitride Ceramic Shapal Hi-M Soft	0.4998	0.3333
Graphite Grade POCO AXF-5Q	1.0000	1.0000

Table D.10: Channel material properties weighted normalized values

Material name	Electrical resistivity (·cm) at 25 degrees	Thermal Conductivity (W/m·K) at 25 degrees
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.0050	0.0000
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.0050	0.0259
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.0050	0.0328
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.0005	0.0259
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.0050	0.2000
Alumnum Nitride Ceramic Shapal Hi-M Soft	0.0500	0.1345
Graphite Grade POCO AXF-5Q	0.0000	0.1397

Table D.11: Channel material properties weighted normalized values

Material name	Coefficient of thermal expansion weighted normalized values	Specific heat capacity weighted normalized values
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.0962	0.0350
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.1000	0.0363
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.0051	0.0500
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.0910	0.0413
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.0872	0.0413
Alumnum Nitride Ceramic Shapal Hi-M Soft	0.0372	0.0000
Graphite Grade POCO AXF-5Q	0.0000	0.0275

Table D.12: Channel material properties weighted normalized values

Material name	Maximum use temp inert weighted normalized values	Density weighted normalized values
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.0000	0.0380
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.0000	0.0426
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.1050	0.0000
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.0225	0.0449
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.1500	0.0472
Aluminium Nitride Ceramic Shapal Hi-M Soft	0.1350	0.0247
Graphite Grade POCO AXF-5Q	0.1500	0.0500

Table D.13: Channel material properties weighted normalized values

Material Name	Material Cost (£/in ³)	Machining Ease (1–10, descending)
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M	0.2200	0.1167
Boron Nitride-Silica Composite (BN-SiO ₂) Grade M26	0.2200	0.1167
Alumina (Al ₂ O ₃) CeramAlox 99.7	0.2471	0.0000
Hexagonal Boron Nitride Composite (h-BN) Grade HP	0.1072	0.0833
Hexagonal Boron Nitride Composite (h-BN) Grade AX05	0.0000	0.0833
Aluminium Nitride Ceramic Shapal Hi-M Soft	0.1249	0.0500
Graphite Grade POCO AXF-5Q	0.2500	0.1500